

When Do Native Speech-Language Models Beat Cascades?

Esther Lee Yi Shan · Nanyang Technological University, Singapore

URECA Supervisor: Asst Prof Dmitrii Ustiugov · Collaborating
Supervisors: Wenyan Chen, Hongrui Liu

Motivation

- Conversational AI has **transitioned from text-only models to native, end-to-end speech-to-speech (S2S) frameworks**
- Modern voice agents integrate acoustic and linguistic features directly, bringing human-like emotional expressiveness (EQ) and natural dialog flows to users
- Commercial and open-source models have **achieved massive scale** (e.g., Doubao surpassing 200 million monthly active users), transforming telecommunications, customer support, and direct human-machine collaboration.

State-Of-The-Art Voice Agents



ChatGPT-4o

- Native Voice-Voice
- Full-Duplex Chat
- Native Tool-Use



Doubao

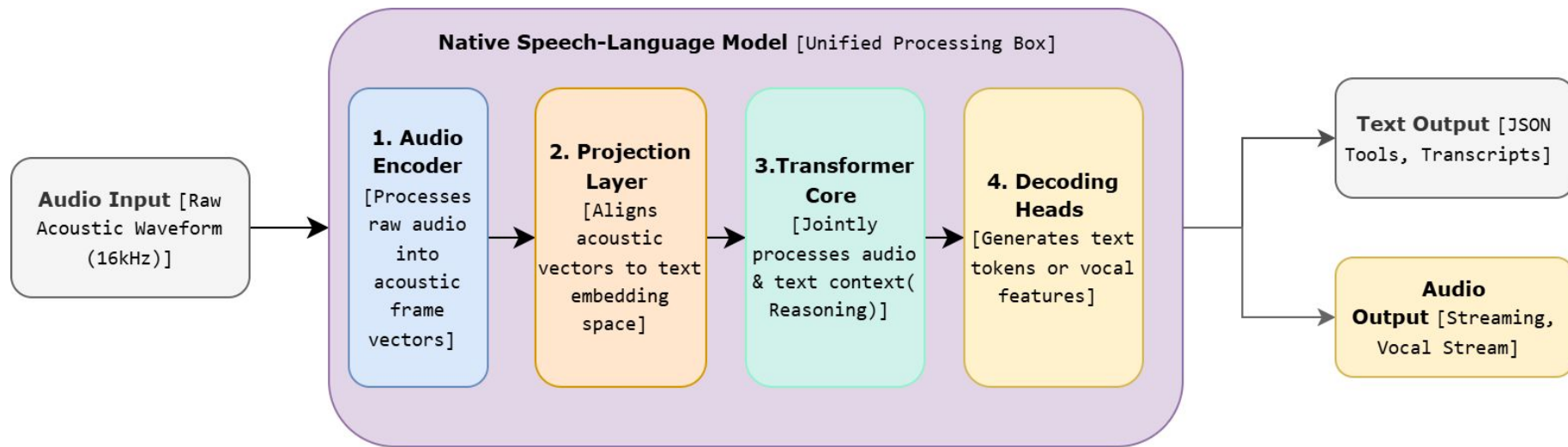
- Over 200M Users
- High-EQ Expressiveness
- Ultra-Low Latency



Qwen-Omni

- Open-Source SOTA
- “Think-Talker” MoE
- Dynamic Reasoning

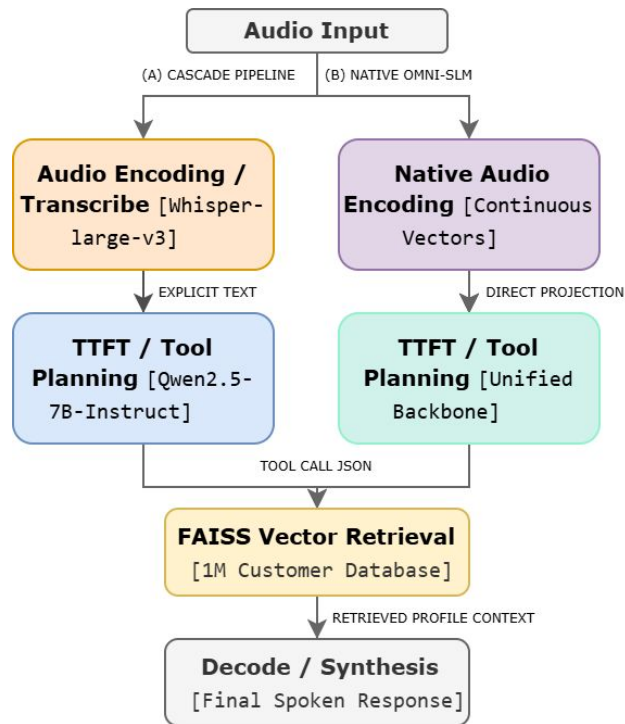
How Native Speech-Language Models Work



- Raw wave signals bypass text transcriptions (ASR) entirely, mapping acoustic frames straight into the model's linguistic embedding space.

Problem

- Traditional speech assistants rely on cascaded pipelines (ASR $\rightarrow$ LLM)
- These modular architectures suffer from severe error propagation (A single typo in the speech step ruins the reasoning step)
- Chaining multiple models causes high latency
- Direct speech models promise to fix this but their limits are untested



Production Requirements

To work in the real world, a voice assistant must master 3 areas:

- Tone and Meaning
 - Understanding both spoken words and voice emotions/nuances.
- Perfect Accuracy
 - Parsing phone numbers and IDs without a single typo.
- Fast Responses
 - Keeping delays low enough for a natural, real-time conversation.

Research Question

Can native Omni-SLMs **replace** traditional cascaded pipelines **under these stringent enterprise constraints?**

Experiment Setup

Models Under Test

1. **Cascade Baseline** (ASR +LLM)

- a. Whisper-large-v3
- b. Qwen2.5-7B-Instruct
- c. Served via vLLM token

2. **Qwen3-Omni-30B-AWQ** (Native Omni-SLM)

- a. Alibaba
- b. 31.6B total parameters
- c. 4-bit AWQ quantized

3. **Nemotron-3-Nano-Omni-30B-BF16**

- a. NVIDIA
- b. 31.6B Total, 3.6B active parameters
- c. MoE BF16

Dataset & Infrastructure

Table 1: Acoustic and Demographic Parameters of the $N = 500$ Evaluation Dataset

Acoustic Parameter	Value / Configuration
Sampling Frequency	16.0 kHz
Resolution	16-bit Linear PCM
Channel Mode	Mono
Average Utterance Duration	4.82 s
Spoken Format	Standard E.164 (International)
Stratification Ratio	20 / 80 (Stratified Interleaved)
Unique Speaker Profiles	1 (Default gTTS Voice)

Accuracy & Alignment Results

Table 2: Operational Accuracy, Vector Alignment, and Hallucination Suppression Profiles ($N = 500$)

Model	Tool Prec.	Tool Recall	Tool F1	TP	FP	FN	HR@1	Halluc. Score
Cascade Baseline	0.957	0.890	0.922	89	4	11	0.978	0.116
Qwen3-Omni-30B-AWQ	0.850	1.000	0.919	96	17	0	1.000	0.135
Nemotron-3-Nano-Omni-BF16	1.000	0.970	0.985	97	0	3	0.856	0.009

Table 3: Latency Performance Breakdown and Quantiled Tail Latencies (ms)

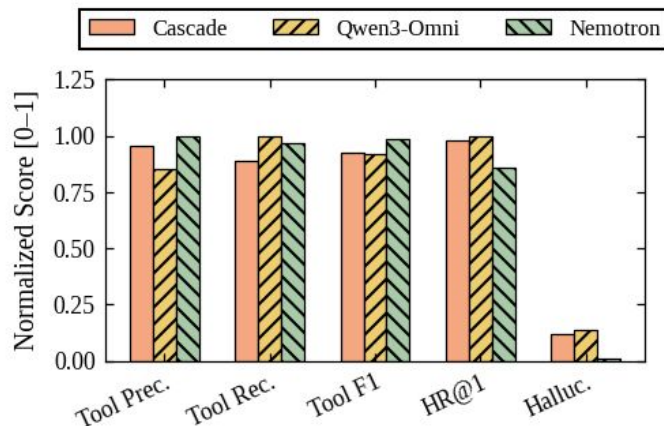
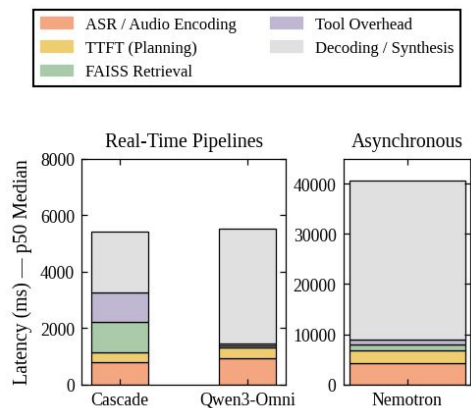
Model	Mean E2E	p50 E2E	p95 E2E	ASR / Enc	TTFT	FAISS	Tool Exec	Decoding
Cascade Baseline	5,529	3,655	14,916	788	364	1,047	1,065	2,137
Qwen3-Omni-30B-AWQ	8,732	6,948	20,381	915	407	67	67	4,056
Nemotron-3-Nano-Omni-BF16	52,593	39,287	132,815	4,290	2,502	1,098	1,116	31,577

Accuracy & Alignment Results

- **Nemotron wins on tool intent**, Achieved an F1-score of 0.985 with 0 false positives.
- **Qwen3-Omni wins on retrieval**, Secured a 1.000 HR@1 database hit rate.
- **Nemotron leads fact grounding**, delivered a near-0 hallucination score (0.009)

Latency Analysis

- Near-zero hallucinations (0.009) and high tool precision (Nemotron) incur a 7x latency penalty (~39.3s), leaving Cascade and quantized models as viable options for real-time customer deployment (<7s)



Where should each architecture be deployed in production?

Cascade Baseline (Whisper-large-v3 + Qwen2.5-7B)

- Should be deployed in **Real-Time Voice Agent Analytics**
- Target Use Case: Live, turn-taking customer calls, where speed is the highest priority
- It has an extremely fast response times (median delay of **3.66 seconds**), fitting safely within natural human conversational limits.
- **However, ASR mistakes pass directly to the LLM with no way to recover. Requires extra text-based guardrails to prevent hallucinations.**

Qwen3-Omni-30B-AWQ (Native Speech Model)

- Should be deployed in **Real-Time Retrieval Agent**
- Target Use Case: Complex, database-reliant lookups where retrieving the correct record is critical.
- It has an flawless vector database alignment (**1.000 HR@1 hit rate**) with highly competitive serving speeds (**median delay of 6.95 seconds**).
- **However, slightly prone to over-triggering tool calls (17 False Positives) and minor quantization-driven hallucinations.**

Nemotron-3-NanoOmni-30B-BF16 (MoE)

- Should be deployed in **Offline Audit & Analytics**
- Target Use Case: Asynchronous post-call processing, call-log validation, and quality assurance auditing.
- It has a near-perfect classification logic (**0.985 F1-score**) and an exceptionally low hallucination rate (**0.009**).
- **However, extremely slow response times (median delay of 39.3 seconds) driven by unquantized Mixture-of-Experts (MoE) routing and memory bottlenecks.**

Key Findings

- **Omni-SLMs are NOT strictly superior**
- **Quantization is essential**
- **Task-specific deployment matters**
- **BERTScore underestimates digit errors**

Conclusion

For production-ready voice assistants, enterprises should adopt a hybrid, task-specific taxonomy:

- **Cascade Pipelines:** Prioritize latency for live, interactive phone routing.
- **AWQ Omni-SLMs:** Prioritize retrieval recall for real-time, data-grounded lookup agents.
- **BF16 Omni-SLMs:** Prioritize grounding accuracy for offline, high-fidelity data auditing and QA.

Future Directions

1. **Noisy Environment Testing:** Assess unified audio encoder performance under real-world distortions.
2. **End-to-End Voice-to-Voice Evaluation:** Benchmark voice-out models that synthesize raw waveforms directly, bypassing text generation.
3. **Entity-Aware Hallucination Metrics:** Combine BERTScore semantic similarity with rule-based string matching for numeric verification.

Thank You!

When Do Native Speech-Language Models Beat Cascades?

Esther Lee Yi Shan · Nanyang Technological University, Singapore

URECA Supervisor: Asst Prof Dmitrii Ustiugov · Collaborating
Supervisors: Wenyan Chen, Hongrui Liu

Background

- Add some sentences to describe the voice agents are important in recent years
- Add some figures to show (examples: Doubao, Qianwen, chatGPT, ...)
- Add a workflow to show users input audio->speech language model->audio/text output



Background

- Conversational AI has **transitioned from text-only models to native, end-to-end speech-to-speech (S2S) frameworks**
- Modern voice agents integrate acoustic and linguistic features directly, bringing human-like emotional expressiveness (EQ) and natural dialog flows to users
- Commercial and open-source models have **achieved massive scale** (e.g., Doubao surpassing 200 million monthly active users), transforming telecommunications, customer support, and direct human-machine collaboration.

State-Of-The-Art Voice Agents



ChatGPT-4

- Native Voice-Voice
- Full-Duplex Chat
- Native Tool-Use



Douba

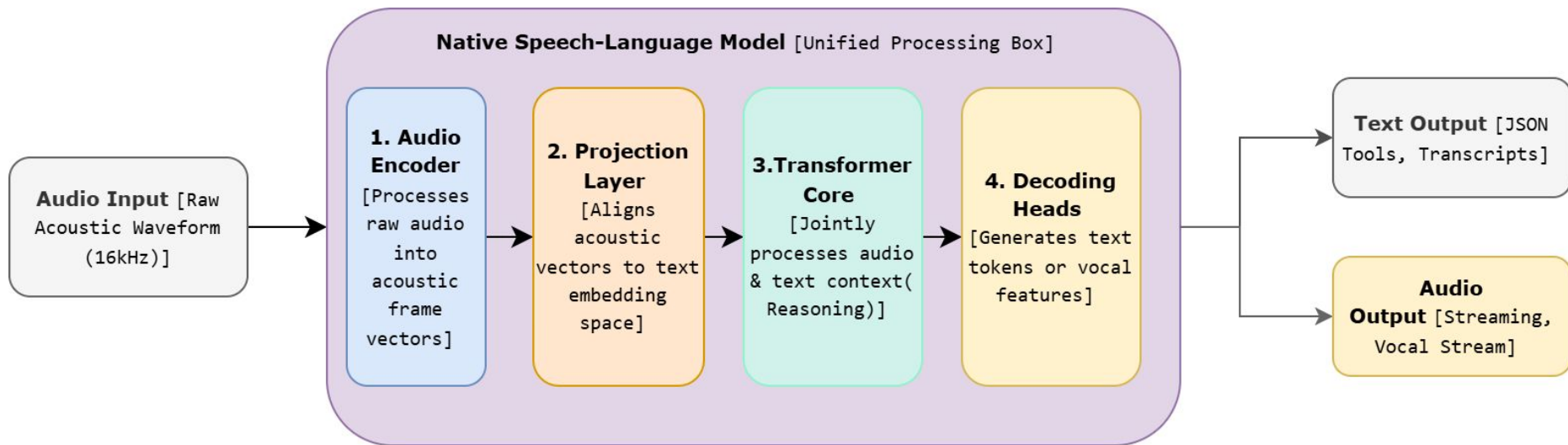
- Over 200M Users
- High-EQ Expressiveness
- Ultra-Low Latency



Qwen-Om

- Open-Source SOTA
- “Think-Talker” MoE
- Dynamic Reasoning

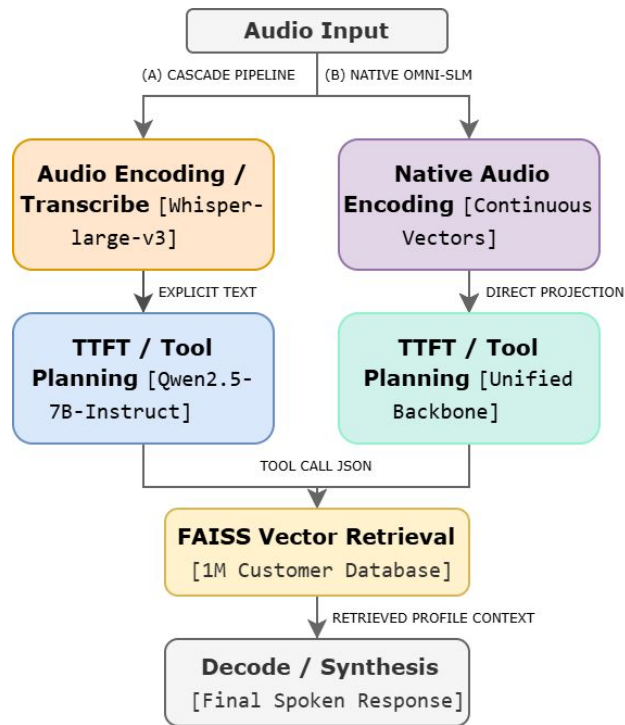
How Native Speech-Language Models Work



- Raw wave signals bypass text transcriptions (ASR) entirely, mapping acoustic frames straight into the model's linguistic embedding space.

Problem

- Traditional speech assistants rely on cascaded pipelines (ASR $\rightarrow$ LLM)
- These modular architectures suffer from severe error propagation (A single typo in the speech step ruins the reasoning step)
- Chaining multiple models causes high latency
- Direct speech models promise to fix this but their limits are untested



Production Requirements

To work in the real world, a voice assistant must master 3 areas:

- Tone and Meaning
 - Understanding both spoken words and voice emotions/nuances.
- Perfect Accuracy
 - Parsing phone numbers and IDs without a single typo.
- Fast Responses
 - Keeping delays low enough for a natural, real-time conversation.

Research Question

Can native Omni-SLMs **replace** traditional cascaded pipelines **under these stringent enterprise constraints?**

Experiment Setup

Models Under Test

1. Cascade Baseline (ASR +LLM)

- a. Whisper-large-v3
- b. Qwen2.5-7B-Instruct
- c. Served via vLLM token

2. Qwen3-Omni-30B-AWQ (Native Omni-SLM)

- a. Alibaba
- b. 31.6B total parameters
- c. 4-bit AWQ quantized

3. Nemotron-3-Nano-Omni-30B-BF16

- a. NVIDIA
- b. 31.6B Total
- c. 3.6B active parameters
- d. MoE BF16

Dataset & Infrastructure

Table 1: Acoustic and Demographic Parameters of the $N = 500$ Evaluation Dataset

Acoustic Parameter	Value / Configuration
Sampling Frequency	16.0 kHz
Resolution	16-bit Linear PCM
Channel Mode	Mono
Average Utterance Duration	4.82 s
Spoken Format	Standard E.164 (International)
Stratification Ratio	20 / 80 (Stratified Interleaved)
Unique Speaker Profiles	1 (Default gTTS Voice)

Accuracy & Alignment Results

Table 2: Operational Accuracy, Vector Alignment, and Hallucination Suppression Profiles ($N = 500$)

Model	Tool Prec.	Tool Recall	Tool F1	TP	FP	FN	HR@1	Halluc. Score
Cascade Baseline	0.957	0.890	0.922	89	4	11	0.978	0.116
Qwen3-Omni-30B-AWQ	0.850	1.000	0.919	96	17	0	1.000	0.135
Nemotron-3-Nano-Omni-BF16	1.000	0.970	0.985	97	0	3	0.856	0.009

Table 3: Latency Performance Breakdown and Quantiled Tail Latencies (ms)

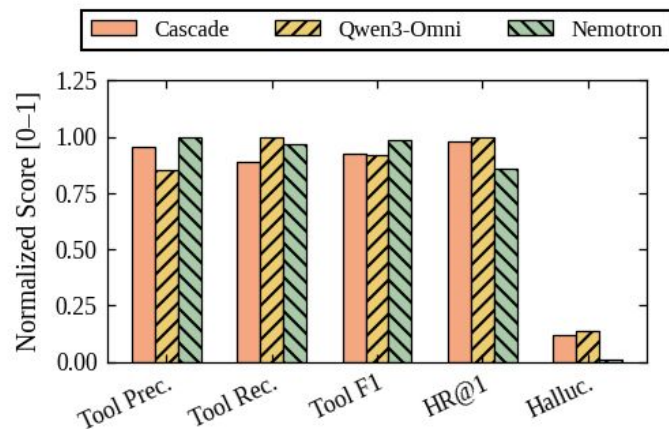
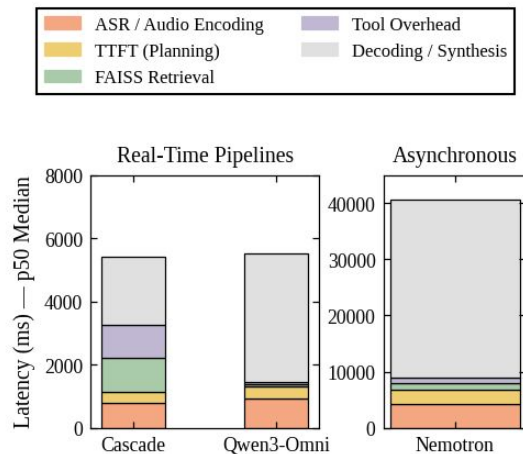
Model	Mean E2E	p50 E2E	p95 E2E	ASR / Enc	TTFT	FAISS	Tool Exec	Decoding
Cascade Baseline	5,529	3,655	14,916	788	364	1,047	1,065	2,137
Qwen3-Omni-30B-AWQ	8,732	6,948	20,381	915	407	67	67	4,056
Nemotron-3-Nano-Omni-BF16	52,593	39,287	132,815	4,290	2,502	1,098	1,116	31,577

Accuracy & Alignment Results

- **Nemotron wins on tool intent**, Achieved an F1-score of 0.985 with 0 false positives.
- **Qwen3-Omni wins on retrieval**, Secured a 1.000 HR@1 database hit rate.
- **Nemotron leads fact grounding**, delivered a near-0 hallucination score (0.009)

Latency Analysis

- Add takeaway of these two figures



Where should each architecture be deployed in production?

Cascade Baseline (Whisper-large-v3 + Qwen2.5-7B)

- Should be deployed in **Real-Time Voice Agent Analytics**
- Target Use Case: Live, turn-taking customer calls, where speed is the highest priority
- It has an extremely fast response times (median delay of **3.66 seconds**), fitting safely within natural human conversational limits.
- **However, ASR mistakes pass directly to the LLM with no way to recover. Requires extra text-based guardrails to prevent hallucinations.**

Qwen3-Omni-30B-AWQ (Native Speech Model)

- Should be deployed in **Real-Time Retrieval Agent**
- Target Use Case: Complex, database-reliant lookups where retrieving the correct record is critical.
- It has an flawless vector database alignment (**1.000 HR@1 hit rate**) with highly competitive serving speeds (**median delay of 6.95 seconds**).
- **However, slightly prone to over-triggering tool calls (17 False Positives) and minor quantization-driven hallucinations.**

Nemotron-3-NanoOmni-30B-BF16 (MoE)

- Should be deployed in **Offline Audit & Analytics**
- Target Use Case: Asynchronous post-call processing, call-log validation, and quality assurance auditing.
- It has a near-perfect classification logic (**0.985 F1-score**) and an exceptionally low hallucination rate (**0.009**).
- **However, extremely slow response times (median delay of 39.3 seconds) driven by unquantized Mixture-of-Experts (MoE) routing and memory bottlenecks.**

Key Findings

- **Omni-SLMs are NOT strictly superior**
 - Each architecture occupies a distinct niche. No single model dominates across all dimensions.
- **Quantization is essential**
 - AWQ 4-bit compression shrinks Qwen3-Omni to 17GB while preserving digit-sequence accuracy and enabling sub-7s latency.
- **Task-specific deployment matters**
 - Cascade for live calls · AWQ-Omni for retrieval agents · BF16-Omni for offline auditing.
- **BERTScore underestimates digit errors**
 - A single-digit phone number mismatch scores near-zero hallucination but is a catastrophic retrieval failure in practice.

Conclusion

To build production-ready voice assistants, enterprise architectures must shift from monolithic, single-model deployments to a hybrid, task-specific taxonomy:

- Cascade Pipelines - Deployed for live, highly interactive customer phone routing (prioritizing latency).
- AWQ Omni-SLMs - Deployed for real-time, data-grounded lookup agents (prioritizing retrieval recall).
- BF16 Omni-SLMs - Deployed for offline, high-fidelity data auditing and QA (prioritizing grounding accuracy).

Future Directions

1. Noisy Environment Testing

- Evaluating how unified audio encoders handle real-world distortions..

2. End-to-End Voice-to-Voice Evaluation

- Moving beyond text-generation outputs to benchmark models that synthesize raw speech waveforms directly (voice-out pipelines).

3. Entity-Aware Hallucination Metrics

- Developing evaluation protocols that combine semantic similarity (BERTScore) with rule-based string matching for numeric database records.

Thank You!